

Geospatial Flood Risk Modeling Using Climate Data and Hidden Markov Models in Rwanda

Problem background

Rwanda continues to face flood risk in urban and peri-urban areas, where intense rainfall, steep slopes, informal settlement patterns, poor drainage, and land-cover change combine to increase surface runoff. In Kigali, flood-prone corridors are often located near streams, wetlands, low-lying settlements, and drainage channels, where heavy rainfall can quickly lead to inundation, road disruption, property damage, and public safety concerns.

The project should focus on a clearly defined flood-prone area rather than Rwanda as a whole. For example, the Nyabugogo-Mpazi corridor in Kigali can be used because it is widely known as a flood-prone urban catchment. Other possible study areas include Gatsata, Gikondo, Rugunga, Rwandex, Mulindi, and floodplain areas near the Nyabarongo system. The final study area should be selected based on the availability of rainfall data, official flood-risk polygons, topographic data, and meaningful geospatial features such as rivers, roads, buildings, and public facilities.

The main research problem is not only whether a location is flood-prone. Static flood-risk maps already provide some information about vulnerable zones. The stronger problem is how daily rainfall and geospatial conditions can be used to model changes in flood pressure over time, so that the system can estimate whether an area is likely to remain in a low, moderate, or high flood-risk state after rainfall events.

Problem gap

Existing flood-risk resources can show areas that are generally vulnerable, but they often remain static. They may not show how flood pressure changes from day to day after rainfall. Climate services provide rainfall information, but rainfall values alone do not directly explain how a particular urban catchment moves between different levels of flood pressure. There is, therefore, a gap in integrating daily climate data, spatial exposure indicators, and temporal state modeling into a single reproducible machine learning workflow.

The gap to be addressed is the lack of a localized geospatial ML framework that combines real rainfall data, official flood-risk spatial layers, terrain features, and urban exposure indicators to estimate flood-pressure states over time. The project should not claim to predict confirmed flood events unless verified daily flood-event records are available. A safer and more academically honest framing is that the model estimates derived flood-pressure states, which can later be compared with official flood-risk zones or, where available, incident records.

Existing solutions and limitations

The relevant existing solutions fall into three groups. First, official flood-risk maps and hazard layers identify areas that are vulnerable to flooding, but they are usually spatial snapshots rather than daily temporal models. Second, climate platforms provide rainfall and temperature data, but

they do not always translate daily rainfall accumulation into flood-pressure states for a specific urban corridor. Third, machine learning flood studies often classify flood susceptibility using terrain, land cover, and distance-to-river features, but many do not include a temporal state model that shows how an area moves between low, moderate, and high risk over time.

This project should position itself between these gaps. The aim is not to replace hydrological engineering models or official warning systems. The aim is to build a small, reproducible, data-grounded ML pipeline that shows how rainfall and geospatial features can be converted into daily flood-pressure states and analyzed using a Hidden Markov Model or a Markov-based temporal layer.

Proposed solution

The proposed solution is a geospatial flood-risk modeling system that uses real climate and spatial data to estimate daily flood-pressure states for a selected flood-prone area in Rwanda. The system will divide the selected study area into grid cells. Each grid cell will be linked to rainfall features, terrain features, distance to rivers or drainage channels, road and building exposure, and whether it intersects an official flood-risk polygon.

A supervised geospatial model can first be used to classify or score each grid cell as low, moderate, or high flood pressure based on the daily rainfall and static spatial features. A Hidden Markov Model can then be used as the temporal layer to model how these flood-pressure states evolve over time. This allows the work to move beyond a static flood map and answer a more useful question: given the current flood-pressure state and recent rainfall pattern, what is the likely state in the next time step?

The chatbot or dashboard is not the core contribution. The core contribution is the data pipeline and ML framework that turns real geospatial and rainfall data into a time-aware flood-pressure modeling system.

Recommended study area options

Option	Why it is suitable	Important data needed
Nyabugogo-Mpazi corridor, Kigali	Strong urban flood-risk relevance, steep slopes, drainage pressure, dense settlement and infrastructure exposure.	Rainfall, official flood-risk polygons, DEM, slope, waterways, roads, buildings, drainage-related features.
Gatsata or lower Nyabugogo areas	Known low-lying urban areas with road, river, and settlement exposure.	Rainfall, road network, river proximity, elevation, building density, flood-risk polygons.
Gikondo or Rwandex corridor	Urban drainage and wetland-adjacent flood exposure can be studied with geospatial features.	Rainfall, land cover, roads, buildings, waterways, elevation, official risk layers.
Nyabarongo-adjacent floodplain areas	Relevant for riverine flood exposure, but may need a wider hydrological framing.	Rainfall, river distance, elevation, floodplain boundary, land cover, settlement exposure.

Scope of the study

The study should be limited to one selected flood-prone area in Rwanda. The final area should be small enough to allow grid-based modeling and detailed feature engineering. A grid size between 100 m and 250 m can be used, depending on computational capacity and data resolution. The temporal scope can cover 2018 to 2024 or another period where daily rainfall data is available.

The model should focus on flood-pressure states rather than confirmed disaster events unless a verified flood-event database is available. This distinction is important. A flood-pressure state is a derived condition based on rainfall accumulation, terrain, proximity to rivers, and indicators of exposure. A confirmed flood event requires observed incident data. If such incident data is unavailable, the claims must remain limited to flood-pressure modeling and official flood-risk polygon agreement.

Main objective

To develop and evaluate a geospatial machine learning and Hidden Markov Model framework that uses real rainfall and spatial data to estimate daily flood-pressure states for a selected flood-prone area in Rwanda.

Specific objectives

- To define a flood-prone study area in Rwanda and prepare a grid-based spatial unit for modeling.
- To collect and preprocess real rainfall, terrain, flood-risk, and OpenStreetMap geolocation data for the selected study area.
- To engineer daily rainfall features and static geospatial features that explain flood pressure across grid cells.
- To derive low, moderate, and high flood-pressure labels using rainfall accumulation and official flood-risk spatial information.
- To train and compare baseline and geospatial machine learning models for flood-pressure classification.
- To integrate a Hidden Markov Model or Markov-based temporal layer that estimates transitions between flood-pressure states over a defined time step.
- To evaluate the model using classification metrics, spatial agreement with official flood-risk areas, and temporal transition performance.

Research questions

- How can daily rainfall data and geospatial features be combined to estimate flood-pressure states in a selected flood-prone area in Rwanda?

- Which geospatial features, such as slope, elevation, river proximity, building density, road density, and official flood-risk intersection, contribute most to flood-pressure classification?
- How well does a geospatial ML model classify low, moderate, and high flood-pressure states compared with a rule-based rainfall threshold baseline?
- How does a Hidden Markov Model represent transitions between flood-pressure states over the chosen observation interval?
- How closely do predicted high-pressure areas align with official flood-risk polygons or available incident records?

Datasets within the scope of the study

The project should use real data sources. The prepared data package already supports a workflow built around a Kigali flood-risk corridor. The workflow can be run locally, in Google Colab, or in Kaggle to download and process the data, where internet access is available.

Dataset	Purpose in the project	Expected role in the pipeline
Meteo Rwanda or ENACTS rainfall data	Provides Rwanda-specific rainfall data. Daily rainfall is preferred for temporal modelling.	Used to compute daily rainfall, 3-day, 7-day, and 14-day rainfall totals.
CHIRPS daily rainfall	Used as a strong fallback if Meteo Rwanda data cannot be downloaded or accessed in time.	Provides long daily rainfall time series for the selected area.
Official flood-risk polygons from Rwanda GeoPortal	Provides official spatial information on flood-risk areas.	Used to create spatial labels and evaluate agreement with predicted high-risk cells.
SRTM elevation data	Provides elevation values for the study area.	Used to derive elevation and slope features.
OpenStreetMap data	Provides rivers, streams, roads, buildings, schools, clinics, hospitals, and other exposure features.	Used to calculate distance to rivers, road density, building density, and exposure indicators.
Optional flood incident records	Can strengthen validation if available from disaster reports, city records, or public incident sources.	Used only if records are credible, dated, and geolocated.

Expected input data structure

The final modeling table should have one row per grid cell per day. This structure allows rainfall to change over time while terrain, river distance, road density, building density, and official flood-risk intersection remain attached to each spatial unit.

Column group	Example columns	Description
Identifiers	date, grid_id, latitude, longitude	Identify the day and spatial unit being modelled.
Rainfall features	rainfall_mm, rain_3d_mm, rain_7d_mm, rain_14d_mm	Capture recent rainfall accumulation and short-term wetness pressure.

Terrain features	elevation_m, slope_deg	Capture how water is likely to move or collect across the area.
Hydrology features	distance_to_river_m, distance_to_stream_m	Capture exposure to rivers, streams, and drainage channels.
Urban exposure features	road_density, building_density, facility_count	Capture the concentration of exposed infrastructure and settlement.
Official risk feature	intersects_flood_risk_polygon	Shows whether the grid cell overlaps with an official flood-risk area.
Target label	flood_pressure_state	Derived label: low, moderate, or high.

Example modeling table

date	grid_id	rainfall_m m	rain_7d_m m	slope_deg	dist_river_m	risk_poly	state
2021-04-01	G001	12.4	46.2	18.5	72	1	moderate
2021-04-02	G001	34.8	78.6	18.5	72	1	high
2021-04-01	G015	5.1	22.7	6.3	410	0	low
2021-04-02	G015	18.0	39.5	6.3	410	0	moderate

Geolocation data collection script

The prepared package includes a script that queries OpenStreetMap through Nominatim and Overpass. Nominatim is used to locate the study area boundary or named places. Overpass is used to download geospatial features such as waterways, roads, buildings, health facilities, schools, and other public infrastructure within the selected area. These features are then converted into distances, densities, and exposure counts for each grid cell.

The geolocation script should be used to produce reproducible spatial layers. The important output files include an area-of-interest file, a grid file, waterways, roads, buildings, facilities, and a final grid feature table. These files are then joined with rainfall and terrain features before model training.

ML framework and pipeline layers

The framework should be built in layers so that the role of each dataset and model is clear.

Layer	What happens in the layer	Output
Layer 1: Study area and grid creation	The selected area is converted into a grid of spatial cells.	grid_id, geometry, centroid coordinates.

Layer 2: Climate feature engineering	Daily rainfall is matched to each grid cell and rolling rainfall totals are computed.	rainfall_mm, rain_3d_mm, rain_7d_mm, rain_14d_mm.
Layer 3: Geospatial feature engineering	Elevation, slope, river distance, road density, building density, and flood-risk polygon intersection are computed.	Static spatial features for each grid cell.
Layer 4: Label construction	Flood-pressure states are derived using rainfall accumulation and official flood-risk information.	low, moderate, or high flood-pressure state.
Layer 5: Geospatial ML classification	A baseline model and ML model classify flood-pressure states.	Predicted flood-pressure state and probability.
Layer 6: HMM temporal modeling	State transitions are estimated over the selected time step.	Transition matrix and next-step state probabilities.
Layer 7: Evaluation and mapping	Predictions are compared with baselines, official flood-risk polygons, and any available incident records.	Metrics, maps, and interpretation.

How the Hidden Markov Model should be used

The Hidden Markov Model should be used as the temporal layer. The hidden states represent the underlying flood-pressure condition of a grid cell: low, moderate, or high. The observed variables are rainfall totals, slope, elevation, distance to river, building density, road density, and flood-risk polygon intersection. The model learns how flood-pressure states transition from one time step to the next.

The time step must be clearly defined. If daily rainfall data is used, then the transition matrix represents one-day transitions. For example, the probability of moving from moderate pressure today to high pressure tomorrow is a daily transition probability. Multi-day forecasts can be estimated by multiplying the transition matrix by itself for the required number of days. If weekly data is used, then the transition matrix represents weekly transitions and the model should be described as a weekly tendency model rather than a daily warning model.

A daily time step is preferred because flood pressure can change quickly after rainfall. If only weekly rainfall summaries are used, the claims must be reduced because weekly modeling cannot support short-term flood warning.

Baselines and Model Comparison

The model should be compared against simple baselines. This is important because the HMM and geospatial ML model must show value beyond basic rainfall thresholds or static flood-risk maps.

Model or baseline	Purpose	Expected comparison
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Rainfall threshold baseline	Classifies flood pressure using fixed rainfall totals only.	Tests whether geospatial features improve over rainfall alone.
Static flood-risk polygon baseline	Classifies cells as risky if they intersect official flood-risk polygons.	Tests whether temporal rainfall features add value beyond static risk maps.
Logistic Regression or Decision Tree	Simple interpretable ML baseline.	Shows whether a basic ML model can classify pressure states.
Random Forest or XGBoost	Main geospatial ML model.	Tests whether nonlinear feature interactions improve classification.
HMM temporal layer	Models state transitions over time.	Tests whether temporal smoothing and transition probabilities improve stability and next-step prediction.

Evaluation Metrics

The project should report both classification and spatial evaluation metrics. If verified incident records are unavailable, the evaluation should be described as agreement with official flood-risk polygons and derived flood-pressure labels, not confirmed flood-event prediction.

Evaluation area	Metrics	Interpretation
Classification performance	accuracy, macro F1-score, balanced accuracy, precision, recall, confusion matrix	Shows how well low, moderate, and high states are classified.
High-risk detection	recall for high-pressure state, precision for high-pressure state	Important because missing high-pressure areas is more serious than over-warning.
Spatial agreement	overlap with official flood-risk polygons, percentage of predicted high-pressure cells inside official risk zones	Shows whether predicted risk aligns with official spatial risk information.
Temporal performance	next-step accuracy, transition matrix interpretation, state sequence stability	Shows whether the HMM captures movement between states over time.
Baseline improvement	difference in F1-score, recall, and high-risk detection compared with threshold and static polygon baselines	Shows whether the proposed framework adds value.

Expected Outputs

- A cleaned grid-cell-by-day dataset for the selected flood-prone area.
- A geospatial feature table containing rainfall, terrain, river distance, road density, building density, and official flood-risk intersection variables.
- A baseline rainfall-threshold flood-pressure classifier.

- A geospatial ML classifier for low, moderate, and high flood-pressure states.
- A Hidden Markov Model transition matrix showing how flood-pressure states change over the selected time step.
- Maps showing predicted high-pressure cells and their overlap with official flood-risk polygons.
- A short dashboard or notebook output that allows the model results to be inspected by date and location.

Important Limitations

The project must clearly separate flood-pressure modeling from confirmed flood-event prediction. If verified flood incident records are not available, the model should not claim to predict actual floods. It should claim to estimate flood-pressure states using rainfall and geospatial risk indicators. The official flood-risk polygons can be used for spatial validation, but they are not the same as daily flood-event records.

The resolution of the rainfall data should also be discussed. CHIRPS provides useful daily rainfall coverage, but it may not capture very localized urban storms as precisely as ground weather stations. The model should therefore be presented as a decision-support and risk-screening framework, not as an official early-warning system.

Some research sources to use but not limited to it:

Mind'je, R., Li, L., Kayumba, P. M., Mindje, M., Ali, S., & Umugwaneza, A. (2021). Integrated Geospatial Analysis and Hydrological Modeling for Peak Flow and Volume Simulation in Rwanda. *Water*, 13(20), 2926. <https://doi.org/10.3390/w13202926>

<https://www.tandfonline.com/doi/abs/10.1080/1573062X.2026.2651433>

<https://www.tandfonline.com/doi/full/10.1080/19475705.2024.2445631>

<https://www.proquest.com/openview/928071fedfe3006e0ce8e20c6bd75b05/1?pq-origsite=gscholar&cbl=18750&diss=y>

<https://webthesis.biblio.polito.it/30042/>

<https://www.mdpi.com/2624-831X/7/1/5>

https://scholar.google.com/scholar?hl=en&as_sdt=0%2C5&q=How+does+a+Hidden+Markov+Model+represent+transitions+between+flood-pressure+states+over+the+chosen+observation+interval%3F&btnG=

<https://www.mdpi.com/2073-4441/11/11/2328>

https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5903444

<https://onlinelibrary.wiley.com/doi/full/10.1111/issr.70005>

<https://tesidottorato.depositolegale.it/handle/20.500.14242/92554>

https://www.igem.qld.gov.au/sites/default/files/2022-10/PROTECTED%20SEQ%20Rainfall%20and%20Flooding%20Reviewreduced_0.pdf

<https://openscholar.uga.edu/record/12058?v=pdf>

<https://www.proquest.com/openview/77b8ed61a029852191ccc0ba9b8d64e4/1?pq-origsite=gscholar&cbl=18750&diss=y>

<https://pmc.ncbi.nlm.nih.gov/articles/PMC12616095/>

<https://www.tandfonline.com/doi/abs/10.1080/19401493.2025.2454258>

https://cartographymaster.eu/wp-content/theses/2021_McCall_Thesis.pdf

<https://www.proquest.com/openview/3a72903f0af192d4100ce610947948e2/1?pq-origsite=gscholar&cbl=18750&diss=y>

<https://www.newtimes.co.rw/article/33240/news/security/kigali-steps-up-drainage-patrols-after-floods-crime-incidents>

https://www.minema.gov.rw/fileadmin/user_upload/Minema/Publications/Disaster_Effects/Annual_Disaster_Effects_Report_2023.pdf

Use other data and research from METEO Rwanda, MINEMA, and other government related reports/studies and East African countries as well